TaxaLoom: interactive exploration of sample-level taxonomic abundance profiles in microbiome datasets

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Abstract

Oui.

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1 Background

High-throughput sequencing has enabled the characterization of complex microbial communities across biological, clinical, and environmental contexts. Taxonomic profiling commonly relies on marker-gene amplicon sequencing or shotgun metagenomics [1, 2]. Marker-gene approaches target taxonomically informative regions, most commonly the 16S rRNA gene for bacterial and archaeal communities, to identify operational taxonomic units (OTUs). In contrast, shotgun metagenomics sequences DNA without targeting specific genomic regions, enabling broader taxonomic and functional characterization [1–3]. Metatranscriptomic sequencing similarly profiles microbial communities from RNA, providing information on transcriptionally active microorganisms and their expressed functions [3, 4]. Despite their methodological differences, these approaches can generate taxonomic abundance profiles describing the microorganisms detected across multiple samples.

Microbial taxa are organized within a hierarchical taxonomy, represented as a branching tree [5], in which each organism belongs to increasingly broader taxonomic groups, from species and genera to families, orders, classes, phyla, and domains [6]. Consequently, exploring microbial abundance data requires navigating this hierarchy while simultaneously considering abundance patterns across taxa and samples. As the number of taxonomic features, the depth of the hierarchy, and the number of samples increase, exploring and visualizing these different dimensions becomes increasingly challenging [7].

The interpretation of microbial abundance profiles depends not only on taxonomic abundances but also on information associated with the samples themselves. Biological and clinical metadata have been widely used to investigate microbiome-host associations, notably through large-scale initiatives such as the Human Microbiome Project (HMP) [8, 9], with associations reported for type 2 diabetes [10] and cancer immunotherapy response [11, 12]. Conversely, technical metadata can help identify variation associated with sample collection, processing, library preparation, or sequencing rather than biological differences. For example, microbial contaminants can vary between laboratories, reagents, and reagent batches [13, 14], an issue of particular relevance when microbial reads represent only a small fraction of host-dominated sequencing data [15]. Joint exploration of taxonomic abundances and sample metadata can therefore help investigate biological or clinical phenotypes while also identifying patterns associated with technical variables.

The exploration of these abundance profiles is further complicated by intrinsic properties of microbiome sequencing data, including differences in sequencing depth across samples, sparsity, and compositionality [16–18]. Numerous normalization and transformation strategies have been proposed to account for these properties, and benchmarking studies have shown that the selected preprocessing strategy can influence downstream analyses, including differential abundance testing, taxon–metadata associations, and phenotype prediction [16, 18–21]. The relative performance of these approaches depends on both the characteristics of the dataset and the objective of the analysis, with no single strategy being optimal in every setting. Examining abundance profiles under different preprocessing strategies can therefore provide complementary

views of the same dataset and help identify patterns that are sensitive to the selected method.

Several tools have been developed to facilitate the visualization and exploration of microbiome taxonomic abundance data (Table 1). Krona uses an interactive sunburst chart to navigate hierarchical taxonomic profiles [22], Metacoder represents taxonomic abundances and differences between groups using heat trees [23], and Pavian uses Sankey diagrams to represent the distribution of reads across taxonomic levels, complemented by heatmaps for multi-sample comparisons [24]. Other tools integrate multiple visualization techniques within broader microbiome analysis environments. Shiny-phyloseq provides bar plots, heatmaps, phylogenetic trees, ordination plots, and interactive network representations [25]. QIIME2 View provides browser-based exploration of QIIME 2 outputs through interactive visualizations including taxonomic bar plots at different taxonomic levels, PCoA ordination plots, alpha-diversity raincloud plots, and summary statistics [26]. MicrobiomeAnalyst combines statistical and preprocessing methods with interactive stacked bar plots, heatmaps, and other representations for microbiome data exploration [27]. Metaviz combines its FacetZoom representation for navigating the taxonomic hierarchy with coordinated heatmaps, stacked bar plots, and other sample-level representations linked through interactive brushing [7]. Snowflake represents microbiome abundance tables as multivariate bipartite graphs connecting samples and OTUs, while incorporating taxonomic annotations and sample metadata and preserving individual microbial features [28]. These tools provide complementary approaches to microbiome visualization but differ in how taxonomic structure, individual samples, associated metadata, and preprocessing choices can be explored together.

Here, we present TaxaLoom, a web-based application for the interactive exploration of sample-level taxonomic abundance profiles. TaxaLoom combines taxonomic hierarchy navigation, sample metadata, synchronized visualizations, and the comparison of preprocessing strategies within a common interface. Although developed using shotgun sequencing data, TaxaLoom can be applied to taxonomic abundance profiles provided as raw count data, independently of the sequencing approach used to generate them. TaxaLoom thereby facilitates the exploration of abundance patterns across taxa, samples, and associated metadata.

2 Implementation

2.1 Technical implementation

2.2 TaxaLoom workflow and input data

TaxaLoom is designed for the interactive exploration of taxonomic abundance profiles together with sample-associated metadata. The application requires an abundance matrix containing taxon abundances across samples, the corresponding taxonomic information, and sample metadata. Taxonomic information provides the hierarchical relationships between detected taxa, whereas metadata can include categorical or quantitative variables describing biological, clinical, or technical characteristics of the samples. **Alternatively, TaxaLoom can take the "BIOM" as input. This**

Table 1: Comparison of TaxaLoom with existing tools for the visualization and exploration of microbiome taxonomic abundance data. Features refer to functionalities described in the corresponding publications.

Feature	TaxaLoom	QIIME2 view [26]	Pavian [24]	Metacoder [23]	MicrobiomeAnalyst [27]	Shiny-phyloseq [25]	Krona [22]	Metaviz [7]	Snowflake [28]
Interface/access mode	Web application	Web application	R/Shiny; web application	R package	Web application	R/Shiny	Browser-based	Web application + R session	R package
Input type	Tabular/BIOM	QIIME2 output	Tabular	Tabular	BIOM; tabular	BIOM	Tabular	Tabular	Tabular
Interactive visualization	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Taxonomic hierarchy visualization	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Limited
Interactive taxonomic filtering	Yes	No	Yes	No	No	No	Limited	Yes	No
Simultaneous visualization of multiple samples	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Sample metadata integration	Yes	Limited	No	Yes	Yes	Yes	No	Yes	Yes
Synchronized interactive views	Yes	No	No	No	No	No	No	Yes	Yes
Built-in normalization	Yes	No	No	Yes	Yes	No	No	No	No
Integrated comparison of normalization methods	Yes	No	No	No	No	No	No	No	No

Yes, functionality available; No, functionality not identified in the corresponding publication or online tool; Limited, functionality available only for a restricted subset of the described feature.

format is widely used in metagenomic dataset, and contains the same information as both previously mentionned matrices [29].

The workflow combines these data into linked representations of the taxonomic structure, sample-level abundance profiles, and sample metadata. Users can navigate between taxonomic levels, restrict the visualization to selected taxa or samples, compare abundance profiles according to metadata variables, and apply different normalization or transformation methods. The data corresponding to the current selection can subsequently be displayed in tabular form and exported for further analysis.

2.3 Interactive visualization

Taxonomic relationships are represented as an interactive tree that allows users to navigate the hierarchical structure of the microbial community. The displayed taxonomic level can be modified during exploration, while a search function and checkboxes allow specific taxa or branches to be selected or hidden, progressively restricting the visualization to taxa of interest.

Taxonomic abundances are simultaneously represented using parallel coordinates and swarm plots. At a selected taxonomic level, each axis of the parallel-coordinates representation corresponds to a taxonomic node and each line represents an individual sample, allowing abundance profiles to be followed across multiple taxa. The corresponding swarm plots represent individual samples as points and provide a complementary view of the distribution of abundances for each displayed taxon. For both representations, abundances can be displayed using either a linear or logarithmic scale.

Sample metadata can be mapped onto these representations to examine abundance patterns in relation to biological, clinical, or technical variables. Both categorical and quantitative metadata are supported, allowing samples to be coloured according to group membership or a continuous variable.

The parallel coordinates, swarm plots, and accompanying data table are linked to facilitate sample-level exploration. Individual samples can be selected or hidden from the table, with selected samples simultaneously highlighted in the parallel coordinates and swarm plots. The table displays abundance values and imported metadata for the currently displayed samples and taxa and can be exported for subsequent analyses, with the current normalization method applied to the data.

2.4 Data normalization and transformation

Microbial abundance data are characterized by differences in sequencing depth across samples, sparsity, and compositionality, making preprocessing an important consideration when exploring abundance profiles [16–18]. As the choice of normalization or transformation method can influence downstream analyses and the resulting representation of microbial communities [16, 18–21], TaxaLoom provides several preprocessing methods that can be applied directly to the imported abundance data. Following selection of a method, abundance values and associated visualizations are dynamically updated at each selected taxonomic level allowing the effect of preprocessing choices to be examined across the complete dataset.

Available preprocessing approaches include total-sum scaling (TSS), binarisation, $\log_2$ counts per million (log2CPM), relative log expression (RLE) [30], trimmed mean of M-values (TMM) [31], arcsine square-root transformation (AsinSQRT), geometric mean of pairwise ratios (GMPR) [19], cumulative-sum scaling (CSS) [32], centered log-ratio (CLR) [33], robust CLR (rCLR) [34], Blom transformation [35], and non-paranormal (NPN) [36]. A description of each preprocessing method and its implementation in TaxaLoom is provided in Supplementary Note B.1.

To streamline the interactive workflow, TaxaLoom prioritizes preprocessing methods that can be applied directly to the abundance matrix without requiring additional choices from the user. For example, the additive log-ratio (ALR) transformation requires the selection of a reference taxon against which all other taxa are expressed [33]. Because such choices require prior knowledge of the dataset and may influence the resulting representation, preprocessing methods requiring additional user-defined parameters or reference features, such as ALR, were not retained.

To compare the abundance profiles obtained using different preprocessing methods, TaxaLoom computes pairwise Pearson, Spearman, or Kendall correlations between the resulting abundance matrices and represents these similarities as a clustered heatmap. This functionality allows users to assess the agreement between preprocessing approaches and to examine whether observed abundance patterns are sensitive to the selected method. An example comparing preprocessing methods applied to the Cleveland cohort is shown in Supplementary Figure S1.

2.5 Case applications

Two case applications were used to illustrate the exploration of microbial abundance profiles with TaxaLoom. The Cleveland cohort was used to examine sample-level patterns in relation to biological and clinical metadata, whereas TCGA data were used

to investigate patterns associated with technical sequencing variables. The datasets and preprocessing steps used for each application are described below.

Cleveland cohort

The Cleveland cohort, originally described by Lederman et al. [37] and subsequently sequenced and analysed by Nganou-Makamdop et al. [38], comprised 55 HIV-infected individuals receiving cART. Participants were classified as immune responders ($n = 11$) or immune non-responders ($n = 44$) based on CD4 T-cell counts above 500 or below 350 cells/μL, respectively, after at least 24 months of virologic suppression on cART [37]. CD4 T-cell counts were also available for individual samples, providing both a categorical immune response status and a quantitative measure of immune recovery.

Raw sequencing data were retrieved from the Sequence Read Archive (SRA) and reanalysed as described in our previous benchmarking study [39]. Following the filtering procedure described in Supplementary Note B.2, the resulting dataset comprised abundance profiles for 235 genera across the 55 samples.

For the case application, we considered the 29 bacterial genera reported by Nganou-Makamdop et al. [38] as differentially abundant between immune responders and immune non-responders: *Mycobacterium*, *Kocuria*, *Microbacterium*, *Lactococcus*, *Hymenobacter*, *Sphingobacterium*, *Xanthomonas*, *Serratia*, *Klebsiella*, *Achromobacter*, *Comamonas*, *Hydrogenophaga*, *Acidovorax*, *Variovorax*, *Ramlibacter*, *Delftia*, *Roseateles*, *Rhizobacter*, *Rubrivivax*, *Massilia*, *Burkholderia*, *Ralstonia*, *Methylophilus*, *Sphingobium*, *Novosphingobium*, *Ochrobactrum*, *Methylobacterium*, *Brevundimonas*, and *Rickettsia*. To limit the influence of highly sparse taxa, genera were required to be detected in at least 22 of the 55 samples, corresponding to a prevalence of at least 40%. Abundance counts were then normalized using the trimmed mean of M-values (TMM) method, and the remaining genera were tested for differences in normalized abundance between immune responders and immune non-responders using a Wilcoxon rank-sum test. Genera with a nominal $p < 0.05$ were retained for visualization, resulting in 12 genera: *Mycobacterium*, *Lactococcus*, *Xanthomonas*, *Serratia*, *Klebsiella*, *Achromobacter*, *Comamonas*, *Variovorax*, *Delftia*, *Massilia*, *Sphingobium*, and *Methylobacterium*.

TCGA dataset

Microbial abundance profiles and associated metadata were obtained from the reanalysis of 5,734 TCGA whole-genome sequencing samples by Ge et al. [15]. Their study reported substantially fewer microbial signals than previous metagenomic reanalyses of TCGA data [40, 41] and identified several previously reported species as known contaminants rather than tumour-associated microorganisms.

For the technical case application, we focused on 122 TCGA thyroid carcinoma (TCGA-THCA) `Primary Tumor` samples, including 26 samples sequenced at `BI` and 96 at `Harvard_GCC_02`. Available technical metadata included sequencing centre and read length, with read lengths of 51, 101, or 151 bp. Taxonomic reconstruction, sample selection, and abundance filtering are described in Supplementary Note B.2.

For visualization, abundance counts were normalized using the geometric mean of pairwise ratios (GMPR) method. The resulting normalized abundances were displayed on a logarithmic scale in the parallel coordinates and swarm plots to facilitate comparison across the wide range of abundance values.

To initiate the exploration of sequencing-associated microbial patterns, we considered *Escherichia coli* and *Cutibacterium acnes*, which were present in the filtered dataset and had previously been identified as potential contaminants in TCGA by Ge et al. [15]. Additional taxa exhibiting patterns associated with the technical metadata were subsequently identified through interactive exploration of the taxonomic tree, as described in the Results.

3 Results

3.1 Case applications

To illustrate the use of TaxaLoom for exploring biological and technical patterns at the individual-sample level, we applied the tool to two independently generated datasets. First, we revisited the Cleveland HIV cohort to explore microbial genera previously associated with immune response to combined antiretroviral therapy (cART) [38, 39]. We then explored microbial profiles derived from The Cancer Genome Atlas (TCGA) [15] to investigate technical patterns associated with sequencing characteristics and associations between microbial profiles and cancer types.

Exploration of microbial profiles associated with immune response to cART

Despite effective suppression of HIV replication, some individuals receiving cART exhibit incomplete CD4 T-cell recovery. Using TaxaLoom, we explored the sample-level abundance profiles of the 12 selected genera in the Cleveland cohort in relation to immune response status and CD4 T-cell counts (Figure 1).

Samples were initially coloured according to immune response status (Figure 1A). Separately examining immune responders (Figure 1B) and immune non-responders (Figure 1C) revealed substantial variability in abundance profiles within both groups. Among immune non-responders, interactive sample selection highlighted a subset of 22 samples with non-zero *Lactococcus* abundance that shared similar abundance profiles across several of the displayed genera.

As the responder/non-responder classification is based on CD4 T-cell recovery, we further explored normalized CD4 T-cell counts as a quantitative variable (Figure 1D). The previously selected *Lactococcus*-positive subset appeared to have lower CD4 T-cell counts. We subsequently tested this visually identified pattern and found significantly lower normalized CD4 T-cell counts in patients with non-zero *Lactococcus* abundance compared with those in whom *Lactococcus* was not detected (Wilcoxon rank-sum test, $W = 540.5$, $p = 0.002$; Figure S2).

This case study illustrates how the combined exploration of taxonomic abundances and categorical or quantitative metadata can reveal sample-level patterns that may not be apparent from group-level comparisons alone.

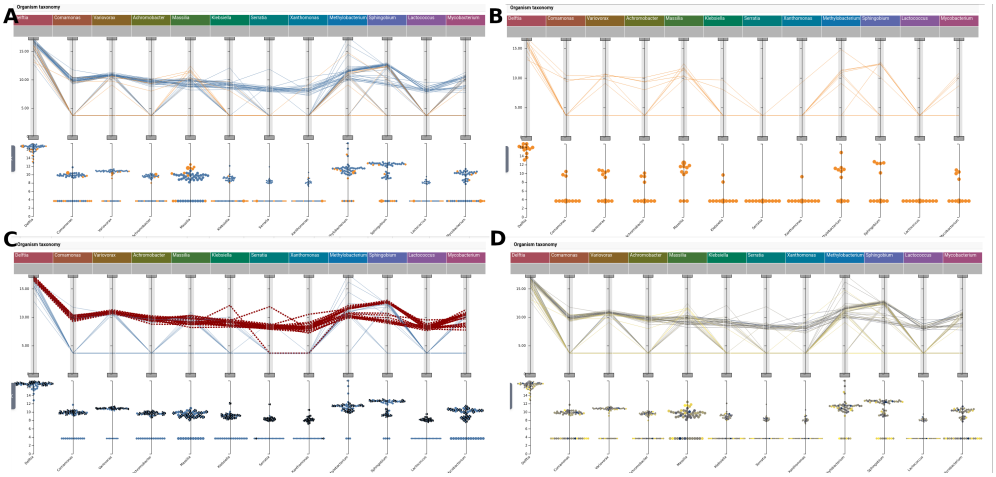


Fig. 1: Exploration of microbial abundance profiles associated with immune response to cART in the Cleveland cohort. TaxaLoom visualization of the 12 selected bacterial genera across 55 samples. (**A**) Samples coloured according to immune response status, with immune responders shown in orange and immune non-responders in blue. (**B**) Visualization restricted to immune responders. (**C**) Visualization restricted to immune non-responders, with the 22 samples with non-zero *Lactococcus* abundance highlighted. (**D**) Samples coloured according to normalized CD4 T-cell counts. Abundance values were normalized using TMM.

3.1.1 Identification of sequencing-associated microbial patterns in TCGA

Using TaxaLoom, we explored TCGA microbial profiles together with technical metadata to investigate sample-level abundance patterns associated with sequencing centre and read length (Figure 2).

We first examined *Escherichia coli* and *Cutibacterium acnes*, the two species present in the dataset that had been identified as potential contaminants by Ge et al. [15]. Exploration of *Enterobacterales* further highlighted *Klebsiella pneumoniae* and *Salmonella enterica* as exhibiting distinct abundance patterns according to sequencing centre (Figure S3).

To identify similar patterns elsewhere in the taxonomy, we progressively filtered the taxonomic tree using TaxaLoom (Figure S4). Starting from the phylum level (Figure S4A), branches were successively removed to focus on *Actinomycetota*, *Bacteroidota*, and *Mycoplasmatota*. The retained branches were then explored at increasingly specific taxonomic levels, from orders (Figure S4B) to genera (Figure S4C), where the visualization was further restricted to *Mycoplasmopsis*, *Plantactinospora*, *Cutibacterium*, and *Phocaeicola*. This progressive filtering ultimately reduced the visualization to species belonging to the selected genera (Figure S4D).

This exploration resulted in nine species retained for the final visualization: *Salmonella enterica*, *Klebsiella pneumoniae*, *Escherichia coli*, *Escherichia albertii*, two

Plantactinospora species, *Cutibacterium acnes*, *Phocaeicola dorei*, and *Mycoplasmopsis bovis*. Their abundance profiles were compared with sequencing centre (Figure 2A) and read length (Figure 2B).

Marked sample-specific patterns were observed across the nine selected species (Figure 2). The clearest pattern was observed for *Escherichia coli*: all samples from `Harvard_GCC_02` had non-zero *E. coli* abundance, whereas *E. coli* was not detected in any sample sequenced at `BI` (Figure 2A). Sequencing centre was completely associated with read length, with all `Harvard_GCC_02` samples having 51-bp reads and `BI` samples having 101 or 151-bp reads (Figure 2B).

The concordance between microbial abundance profiles and these technical variables indicates that technical effects may contribute to the observed patterns. For `Harvard_GCC_02`, the high abundance of *E. coli* and related Enterobacterales could reflect contamination introduced during sample processing. However, the complete association between sequencing centre and read length prevents these effects from being disentangled. The shorter 51-bp reads may also provide lower taxonomic resolution, potentially contributing to classifications among closely related Enterobacterales such as *K. pneumoniae*, *S. enterica*, and *E. albertii*. Conversely, `BI` samples showed higher abundance of *C. acnes*, a common skin-associated contaminant also identified as a potential contaminant in TCGA data by Ge et al. [15]. No clear explanation was identified for the sequencing-associated patterns observed for *M. bovis*, *P. dorei*, and *Plantactinospora* spp.

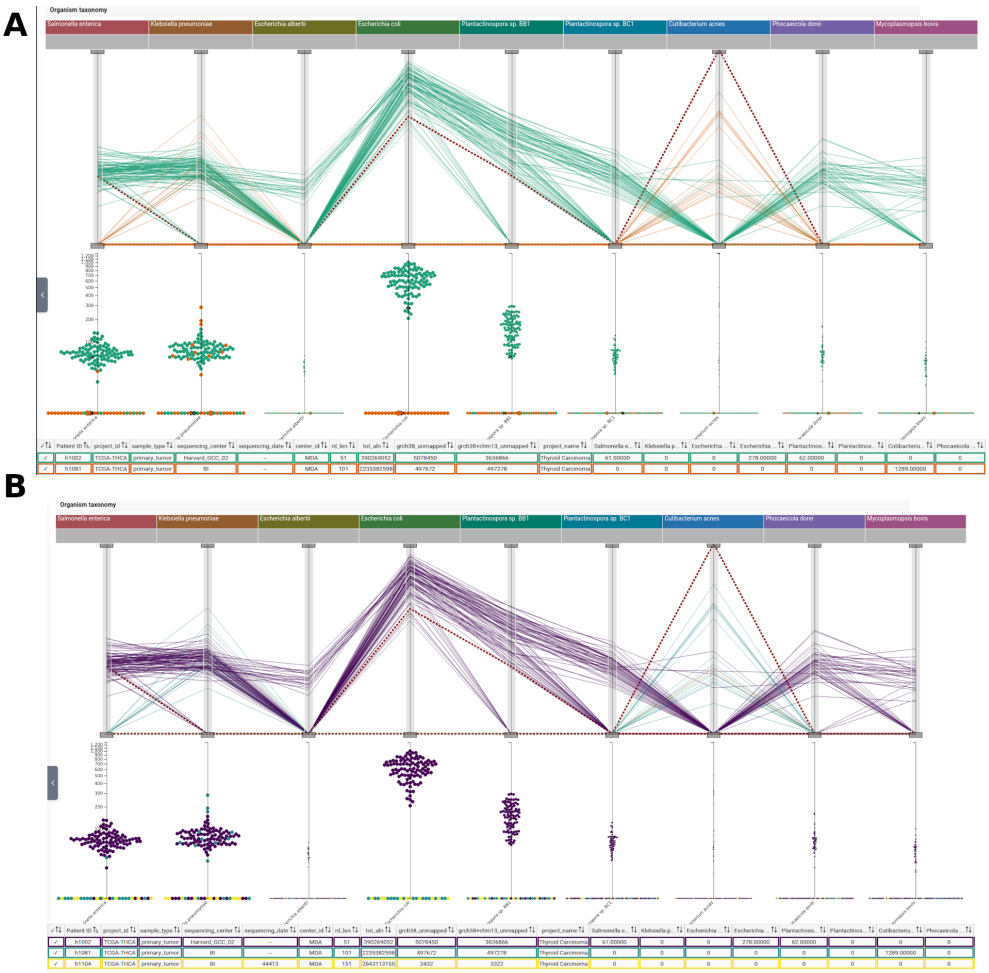


Fig. 2: Sample-level microbial abundance patterns associated with technical variables in TCGA-THCA. TaxaLoom visualization of nine selected microbial species across 122 TCGA-THCA `Primary Tumor` samples. Samples are coloured according to (**A**) sequencing centre, with `Harvard_GCC_02` shown in green and `BI` in orange, and (**B**) read length (51, 101, or 151 bp). Abundances were normalized using GMPR and displayed on a logarithmic scale.

4 Discussion

5 Conclusion

Availability and requirements

Lists the following:

1. Project name: TaxaLoom
2. Project home page: e.g. http://sourceforge.net/projects/mged
3. Operating system(s): e.g. Platform independent Programming language: e.g. Java
4. Other requirements: e.g. Java 1.3.1 or higher, Tomcat 4.0 or higher
5. License: e.g. GNU GPL, FreeBSD etc.
6. Any restrictions to use by non-academics: e.g. licence needed.

Abbreviations

mRNA: Messenger ribonucleic acid
RNA: Ribonucleic acid

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Please refer to Journal-level guidance for any specific requirements.

Declarations

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- Funding
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

Appendix A Supplementary Figures

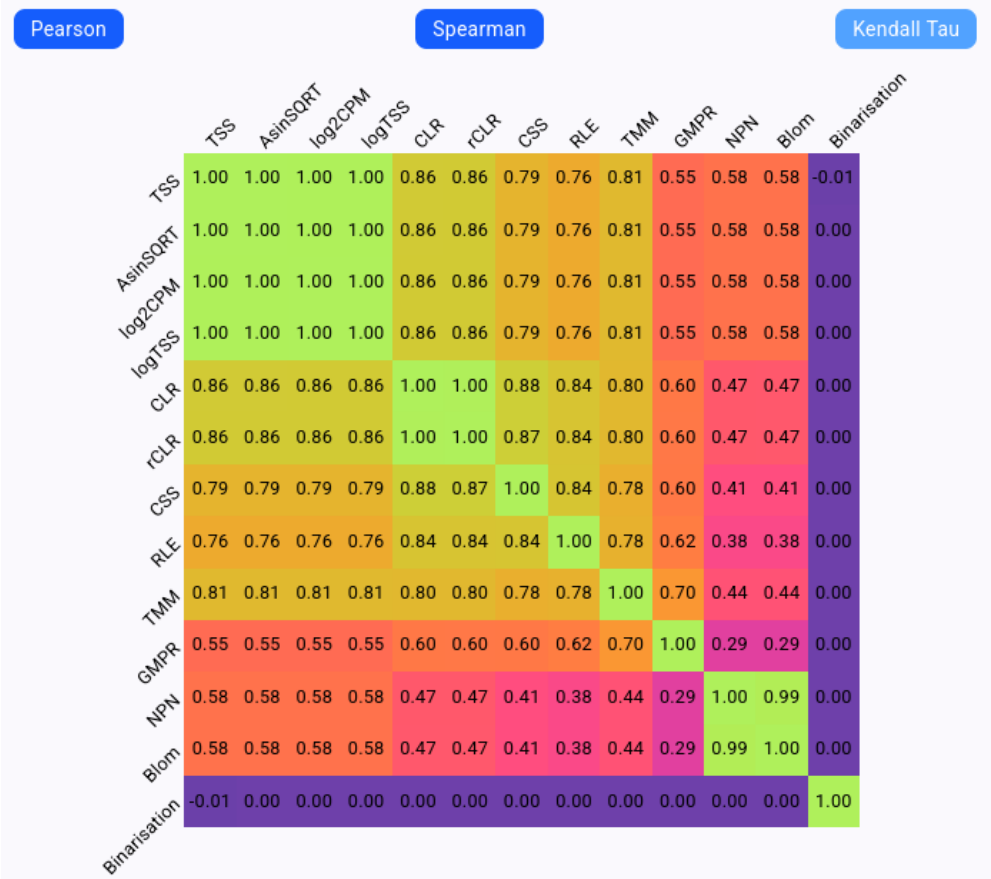


Fig. S1: Comparison of normalization and transformation methods available in TaxaLoom for the Cleveland cohort. Pairwise correlations between abundance matrices obtained using the different normalization and transformation methods available in TaxaLoom. Similarity between matrices was evaluated using Kendall correlation (shown), with Spearman and Peason correlations also available through the interactive interface. Values closer to 1 indicate greater similarity between the abundance profiles obtained with the corresponding methods.

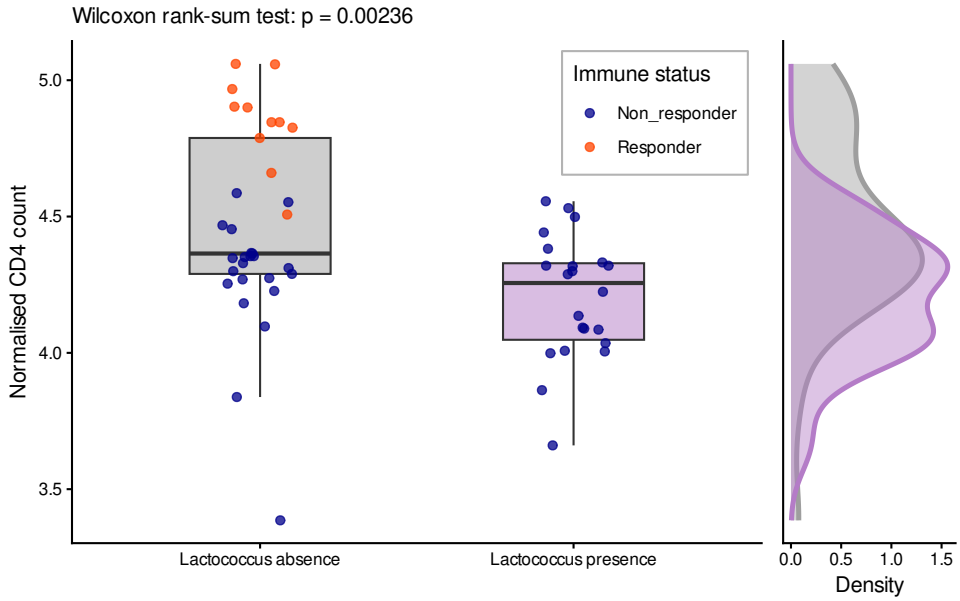


Fig. S2: Boxplots of normalized CD4 T-cell counts according to *Lactococcus* detection in the Cleveland cohort. Distribution of normalized CD4 T-cell counts in samples with and without detected *Lactococcus*. Individual samples are coloured according to immune response status. Boxplots summarize the distribution within each group, with corresponding density distributions shown on the right. Normalized CD4 T-cell counts were significantly lower in samples with non-zero *Lactococcus* abundance (Wilcoxon rank-sum test, $W = 540.5$, $p = 0.00236$).

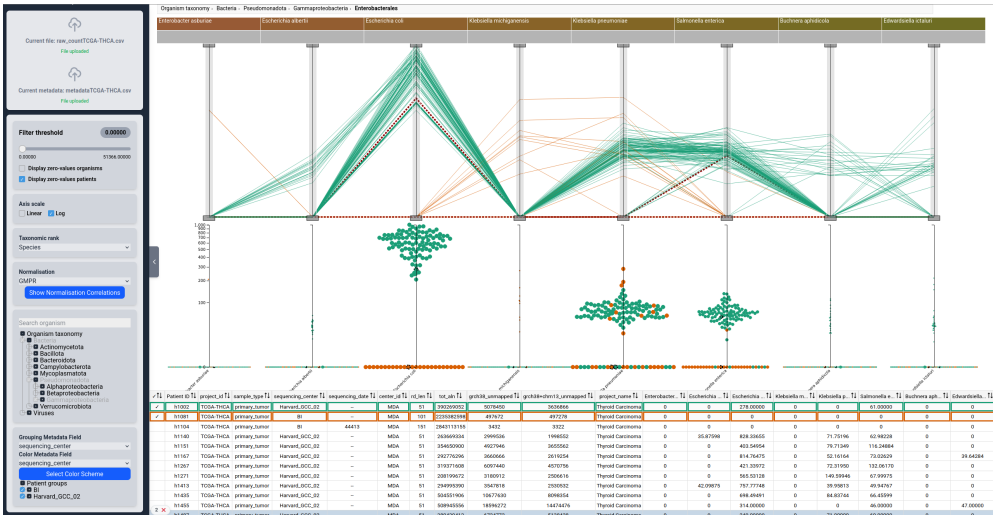


Fig. S3: **Sample-level abundance profiles of Enterobacterales in TCGA-THCA.** TaxaLoom visualization restricted to species belonging to the order *Enterobacterales* in the 98 TCGA-THCA `Primary Tumor` samples. Samples are coloured according to sequencing centre, allowing sequencing-associated abundance patterns to be identified across Enterobacterales species. Abundances were normalized using GMPR and displayed on a logarithmic scale.

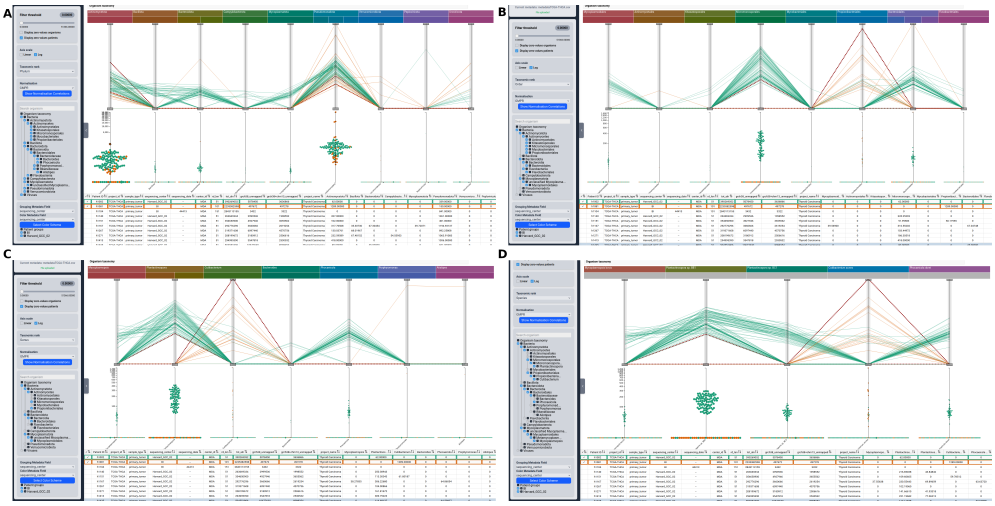


Fig. S4: Progressive taxonomic filtering of TCGA-THCA microbial profiles using TaxaLoom. Taxonomic branches were successively filtered to focus on taxa exhibiting sequencing-associated abundance patterns. (**A**) Initial visualization at the phylum level. Selected phyla were removed to retain *Actinomycetota*, *Bacteroidota*, and *Mycoplasmatota* for further exploration. (**B**) Resulting visualization at the order level, from which *Mycoplasmoidales*, *Micromonosporales*, *Propionibacteriales*, and *Bacteroidales* were retained. (**C**) Genus-level visualization, further restricted to *Mycoplasmopsis*, *Plantactinospora*, *Cutibacterium*, and *Phocaeicola*. (**D**) Final species-level visualization resulting from the successive taxonomic filtering. Abundances were normalized using GMPR and displayed on a logarithmic scale.

Appendix B Supplementary Notes

B.1 Preprocessing methods available in TaxaLoom

Throughout this section, y_{ji} and q_{ji} denotes, respectively the raw and relative abundance of feature i in sample j, and $N_j = \sum_{i=1}^{p} y_{ji}$ denotes the corresponding library size, where p is the total number of features.

Binarisation

Binarisation converts count data into presence-absence values by removing abundance information. Let y_{ji} denote the raw count of feature i in sample j.

The transformation is defined as

$$x_{ji} = \begin{cases} 1 & y_{ji} > 0 \\ 0 & y_{ji} = 0 \end{cases}$$

This transformation encodes only whether a feature is detected in a sample, disregarding quantitative abundance information.

Total Sum Scaling (TSS)

Total sum scaling (TSS) transforms raw counts into relative abundances by normalizing each sample by its sequencing depth. Let y_{ji} denote the raw count of feature i in sample j.

The relative abundance is defined as

$$q_{ji} = \frac{y_{ji}}{N_j}.$$

which rescales each sample to sum to 1.

log2CPM

log2 counts per million (log2CPM) rescales counts by library size and applies a log_2 transformation to stabilize the variance. Let y_{ji} denote the raw count of feature i in sample j.

The transformation is defined as

$$x_{ji} = \log_2 \left(\frac{y_{ji} + 1}{N_j} \times 10^6 \right).$$

where a pseudocount is added to avoid zeros before applying the log transform. It expresses counts as log-scaled counts per million reads, accounting for sequencing depth differences between samples.

TMM Normalisation

TMM (Trimmed Mean of M-values) estimates a sample-specific scaling factor by comparing each sample to a reference sample [30].

In the edgeR implementation, the reference sample r is defined as the sample whose count-per-million upper quartile is closest to the mean upper quartile [29]. For each feature i, an M-value representing the log-fold change between sample j and the reference sample is computed as:

$$M_{ji} = \log_2 \left(\frac{y_{ji}/N_j}{y_{ri}/N_r} \right)$$

where y_{ji} and y_{ri} denote the raw counts for feature i in sample j and reference sample r, respectively, and $N_j = \sum_i y_{ji}$ and $N_r = \sum_i y_{ri}$ are the corresponding library sizes.

An average abundance measure is also calculated as:

$$A_{ji} = \frac{1}{2} \log_2 \left(\frac{y_{ji}}{N_j} \cdot \frac{y_{ri}}{N_r} \right)$$

The M- and A-values are then used to remove features with extreme fold changes or abundance values. Following the default TMM procedure, the 30% highest and lowest M-values and the 5% highest and lowest A-values are excluded [30]. The scaling factor is therefore estimated from features that are neither extreme in fold change nor in abundance.

The weight for feature i and sample j is defined as:

$$w_{ji} = \frac{N_j - y_{ji}}{N_j \cdot y_{ji}} + \frac{N_r - y_{ri}}{N_r \cdot y_{ri}}$$

The TMM scaling factor for sample j relative to the reference is then obtained from the weighted mean of the M-values retained after trimming:

$$\log_2(s_j) = \frac{\sum_{i \in \mathcal{I}} w_{ji} M_{ji}}{\sum_{i \in \mathcal{I}} w_{ji}},$$

where $\mathcal{I}$ denotes the set of features retained after M- and A-value trimming. The resulting normalization factors s_j are rescaled so that their product across samples equals one [29]. As for RLE normalization, the effective library size is then obtained as:

$$N_j^* = N_j s_j,$$

where N_j is the original library size of sample j.

RLE normalisation

Relative log expression (RLE) normalization estimates sample-specific scaling factors using the median-ratio method proposed in [2]. For each feature i, the geometric mean of its counts across all n samples is calculated:

$$g_i = \left(\prod_{j=1}^{n} y_{ji} \right)^{1/n},$$

where y_{ji} denotes the raw count for feature i in sample j. The resulting values g_i define a reference abundance profile against which each sample is compared. The scaling factor s_j for each sample is then calculated as the median of the ratios between its feature counts and the corresponding reference values:

$$f_j = \text{median}\left(\frac{y_{ji}}{g_i}\right).$$

In the edgeR implementation, these factors are rescaled so that their product across samples equals one [29]. The resulting normalization factor s_j is used to calculate the effective library size of sample j:

$$N_j^* = N_j s_j,$$

where N_j is the original library size of sample j.

Blom transform

The Blom transformation converts abundance values into rank-based normal scores [3]. For each feature i, abundance values are first ranked across the n samples. The rank r_{ji} of feature i in sample j is then converted into a value between 0 and 1:

$$q_{ji} = \frac{r_{ji} - 3/8}{n + 1/4},$$

where n is the number of samples. The resulting value q_{ji} is then mapped to the corresponding quantile of the standard normal distribution using its inverse cumulative distribution function:

$$x_{ji} = \Phi^{-1}(q_{ji}),$$

where $\Phi^{-1}(\cdot)$ denotes the inverse cumulative distribution function of a normal distribution with mean 0 and standard deviation 1. The resulting x_{ji} values are therefore rank-based normal scores that approximate a standard normal distribution while preserving the ordering of the original abundance values.

NPN transform

The non-paranormal (NPN) transformation uses a rank-based Gaussianization procedure [19]. For each feature i, abundance values are ranked across the n samples. The rank r_{ji} of feature i in sample j is first mapped to a value between 0 and 1:

$$q_{ji} = \frac{r_{ji}}{n + 1},$$

where n is the number of samples.

These values are then transformed using the inverse cumulative distribution function of the standard normal distribution:

$$x_{ji} = \Phi^{-1}(q_{ji}),$$

yielding rank-based values that approximately follow a normal distribution.

In practice, a truncated version of the NPN transformation was used, as implemented in the *huge* R package [35]. Before applying the inverse normal transformation, values at the lower and upper extremes are constrained to predefined limits. This truncation prevents extreme ranks from producing excessively large positive or negative transformed values.

GMPR transform

GMPR (Geometric Mean of Pairwise Ratios), as implemented in the *MicrobiomeStat* R package, is a size-factor-based normalization method designed for sparse count data [4].

For each pair of samples (j, k), abundance ratios are calculated using only features that are non-zero in both samples. A pairwise ratio is defined as the median of these ratios:

$$r_{jk} = \text{median}_{i \in \mathcal{I}_{jk}} \left(\frac{y_{ji}}{y_{ki}} \right),$$

where $\mathcal{I}_{jk} = \{i : y_{ji} > 0,\ y_{ki} > 0\}$ denotes the set of shared non-zero features between samples j and k.

For each sample j, a sample-specific scaling factor s_j is then calculated as the geometric mean of its pairwise ratios with the other samples:

$$s_j = \left(\prod_{k \neq j} r_{jk} \right)^{\frac{1}{n}},$$

where n is the number of samples. The resulting scaling factor summarizes the abundance of sample j relative to the other samples while reducing the influence of zeros by considering only features detected in both samples.

Normalized abundances are then obtained by dividing the raw counts by the corresponding scaling factor:

$$x_{ji} = \frac{y_{ji}}{s_j}.$$

CSS transform

Cumulative sum scaling (CSS), implemented in the *metagenomeSeq* R package, is a normalization method designed for sparse microbiome count data [26].

CSS determines a percentile of the count distribution from the data, above which highly abundant features are excluded from the calculation of the scaling factor. For each sample j, the scaling factor s_j is then calculated as the cumulative sum of counts up to the selected percentile:

$$s_j = \sum_{i:\, y_{ji} \leq q_l(y_j)} y_{ji},$$

where $q_l(y_j)$ denotes the count value corresponding to the selected percentile in sample j.

Normalized abundances are obtained by dividing each raw count by the corresponding scaling factor:

$$x_{ji} = \frac{y_{ji}}{s_j}.$$

By calculating the scaling factor without the most highly abundant features, CSS reduces their influence on the normalization.

AsinSquareRoot

The arcsine square-root transformation is applied to relative abundance data to stabilize variance, particularly for proportions close to 0 or 1. Let q_{ji} denote the relative abundance of feature i in sample j, obtained after total sum scaling.

The transformation is defined as

$$x_{ji} = \arcsin\left(\sqrt{q_{ji}}\right),$$

The transformation reduces the dependence of the variance on the mean that is characteristic of proportional data, producing values with a more stable variance across the range of relative abundances.

CLR transform

The centered log-ratio (CLR) transformation maps relative abundances to a log-ratio scale relative to the geometric mean of all features within a sample [1]. Let g_j denote the geometric mean of the relative abundances in sample j.

The transformation is defined as:

$$x_{ji} = \ln\left(\frac{q_{ji}}{g_j}\right),$$

where

$$g_j = \left(\prod_{k=1}^{p} q_{jk}\right)^{1/p},$$

and p is the total number of features. The resulting values express the abundance of each feature relative to the geometric mean abundance of all features within the same sample.

rCLR transform

The robust centered log-ratio (rCLR) transformation is a modification of the CLR transformation that accommodates zero values by calculating the geometric mean using only non-zero features [21].

For each sample j, the geometric mean is calculated using only non-zero relative abundances:

$$g_j = \exp\left(\frac{1}{m_j^+}\sum_{k:q_{jk}>0}\ln(q_{jk})\right),$$

where m_j^+ denotes the number of non-zero features in sample j.

The rCLR transformation is then defined as:

$$x_{ji} = \begin{cases} \ln\left(\frac{q_{ji}}{g_j}\right) & q_{ji} > 0, \\ 0 & q_{ji} = 0. \end{cases}$$

The resulting values express each non-zero feature as a log-ratio relative to the geometric mean of the non-zero features within the same sample, while zero values remain zero in the transformed data.

B.2 Microbial abundance filtering

Both datasets used in the case studies were derived from host-dominated sequencing data and we therefore add an additional filtering step before their use in TaxaLoom. The Cleveland cohort was generated from total-RNA sequencing of blood samples, in which human-derived reads represented the majority of the sequencing data [6, 24]. Similarly, the TCGA microbial profiles were obtained from the reanalysis of human whole-genome sequencing (WGS) data [9]. As the microbial signal represents only a small fraction of these datasets, we applied dataset-specific filtering procedures to reduce low-abundance and potentially spurious taxonomic classifications before visualization.

We used a similar strategy for both dataset as they were reanalysed using similar procedure.

Cleveland cohort filtering

Pre-treated raw count data at genus-level were retrieved from [6], and the associated metadata from [24].

First, in order to obtain a normally distributed outcome, the CD4 T-cell count was log-transformed to ease the visualization.

Following taxonomic classification, genera considered biologically implausible in human blood were manually excluded based on their known host range and ecology. This filtering removed 23 genera, including plant-associated viruses and pathogens: *Orthotospovirus* [15], *Albugo* [14], *Phytophthora* [22], *Entyloma* [16], *Sporisorium* [36], *Stenocarpella* [10], *Sclerotinia* [7], *Taphrina* [25], and *Zymoseptoria* [28]. Genera associated with highly specific non-human or environmental niches were also removed, including the bee gut symbiont *Gilliamella* [17]; the haloarchaea *Halobacterium* and *Haloplanus* [8], *Halorubrum* and *Natronomonas* [13], and *Natrinema* [32]; the wood- or tree-associated fungi *Hornodermoporus* [34], *Chrysoporthe* [33], *Immersiporthe* [5], and *Phlebiopsis* [12]; the mushroom-forming genus *Entoloma* [20]; the insect-parasitic nematode *Hexamermis* [18, 27]; and the dung-associated fungus *Sordaria* [23]. Finally, *Sinsheimervirus* was excluded because this genus contains PhiX174, which is commonly used as an Illumina sequencing control [Illumina].

To reduce low-abundance taxonomic assignments, read-count cutoffs ranging from 1 to 50 reads were evaluated independently for each sample. For each cutoff, taxa with fewer reads were considered absent and their counts were set to zero. The resulting relationship between the minimum abundance required for detection and the number of taxa remaining in each sample is shown in Figure S5.

A sample-specific cutoff was selected as the first point at which the mean relative decrease in the number of taxa reached 5% over a sliding window of five consecutive values. This criterion was designed to identify the beginning of a plateau, beyond which increasing the cutoff produced relatively little change in the number of detected taxa, thereby limiting low-abundance classifications potentially resulting from spurious taxonomic assignments.

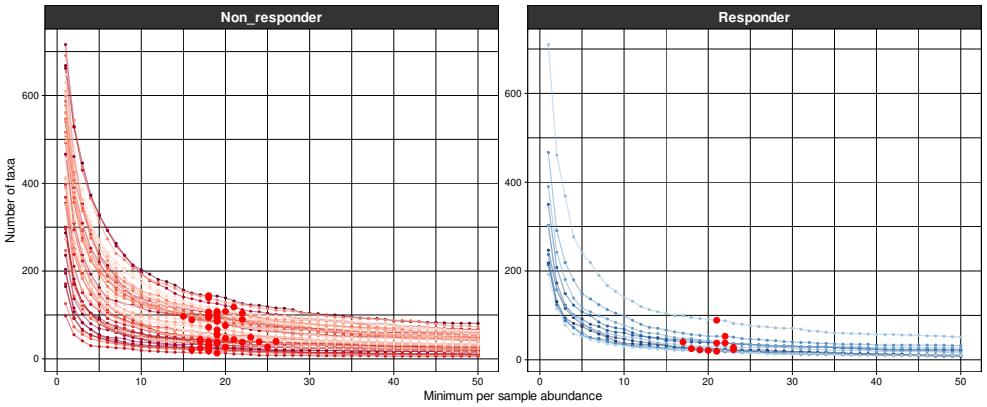


Fig. S5: Sample-specific abundance filtering for the Cleveland cohort. Number of taxa retained in each sample as a function of the minimum read-count threshold required to consider a taxon present. Taxa with counts below the corresponding threshold were considered absent and their counts were set to zero. Red points indicate the threshold selected independently for each sample, defined as the first point at which the mean relative decrease in the number of retained taxa reached 5% over a sliding window of five consecutive thresholds.

TCGA-THCA filtering

Reanalysed TCGA microbial abundance data and associated metadata were obtained from Ge et al. [9]. The complete dataset comprised 5,734 whole-genome sequencing samples and 11,389 detected species. Complete taxonomic lineages were reconstructed from the taxonomic identifiers provided in the dataset using TaxonKit [31]. Samples with fewer than 1,000 total microbial counts were excluded to ensure sufficient microbial signal for subsequent exploration.

For the technical case application, the analysis was restricted to TCGA thyroid carcinoma (TCGA-THCA). The complete TCGA dataset was too large for practical interactive exploration with TaxaLoom, while the objective of this case application

was to investigate technical covariates rather than cancer-specific abundance patterns. TCGA-THCA was selected because it contained a large number of `Primary Tumor` samples from both sequencing centres while remaining sufficiently small for efficient exploration. The resulting dataset comprised 122 `Primary Tumor` samples, including 26 from `BI` and 96 from `Harvard_GCC_02`.

To further reduce low-abundance taxonomic assignments within the retained samples, sample-specific abundance cutoffs ranging from 1 to 250 counts were evaluated independently for each sample. For each cutoff, species with counts below the threshold were considered absent and their counts were set to zero. The selected cutoff corresponded to the first point at which the mean relative decrease in the number of retained taxa fell below 1% over a sliding window of five consecutive cutoff values (Figure S6). The wider range of candidate cutoffs and more stringent plateau criterion compared with the Cleveland cohort were used to accommodate the greater heterogeneity in microbial abundance observed across the TCGA samples.

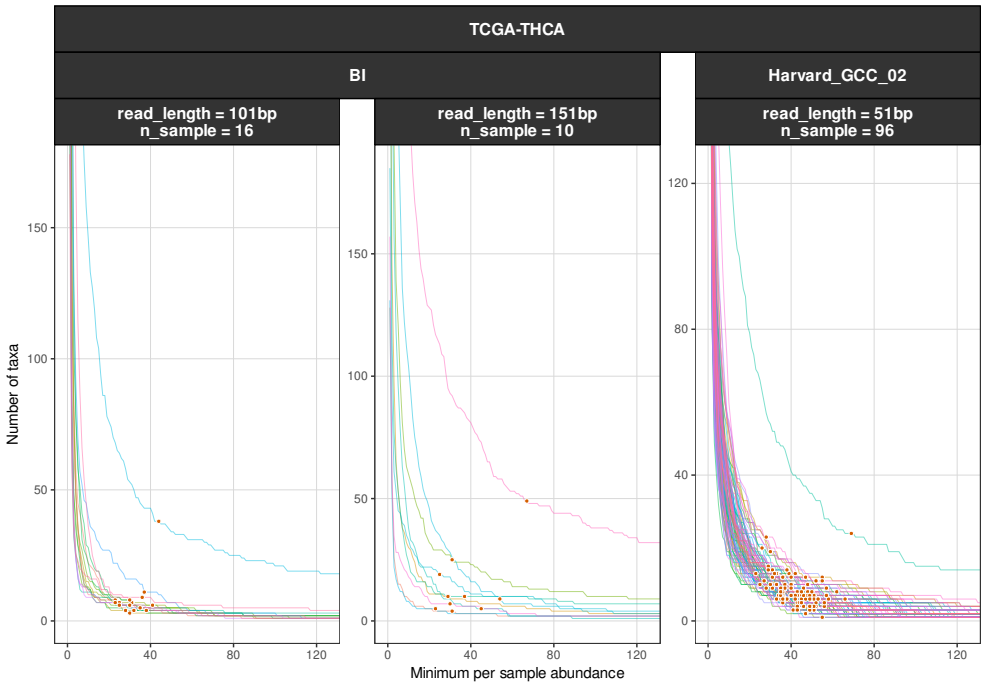


Fig. S6: Sample-specific abundance filtering for the TCGA-THCA, primary tumor samples. Number of taxa retained in each sample as a function of the minimum read-count threshold required to consider a taxon present. Taxa with counts below the corresponding threshold were considered absent and their counts were set to zero. Red points indicate the threshold selected independently for each sample, defined as the first point at which the mean relative decrease in the number of retained taxa reached 1% over a sliding window of five consecutive thresholds.

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